



ORIGINAL PO CASE STUDY | 2013

Hybrid CCTV security. Engineered for every zone.

Verified IP and analog surveillance infrastructure for the Firmenich Dahej SEZ plant - designed, supplied and installed by D-TECH under Purchase Order 3000175931.

D-TECH DELIVERY MODEL

DESIGN → IMPLEMENT → MAINTAIN





ORIGINAL ORDER EVIDENCE

A documented hybrid CCTV deployment for Firmenich Dahej SEZ.

Project facts below are taken directly from Firmenich Purchase Order 3000175931.

2013

order year

Issued 21 June 2013

30

line items

Cameras, network, recording and power

20

cameras

15 analog + 5 IP / PTZ

₹18.52 L

PO value

INR 1,851,800.90 excl. tax

Dahej

project site

Plot Z-10, Dahej SEZ

COMMERCIAL RECORD

Supply, installation and commissioning responsibility in one order.

The PO states that installation charges were included, delivery was for the Dahej site, and the system carried a 12-month manufacturing-defect warranty.

VERIFIED PO SCOPE

A complete camera-to-control-room infrastructure package.

The original order combined field devices, long-distance transmission, recording, display, rack, power protection and installation.

15

Analog IR cameras

IR LAB CIR IH86FV with 15 outdoor housings.

3

Fixed IP cameras

Outdoor IP66-rated AXIS M1113-E with PoE midspans.

2

Network PTZ cameras

AXIS Q6032-E PTZ dome cameras with wall brackets.

9 km

Video cable

RG11 coax for distributed analog camera connectivity.

2 km

Single-mode fiber

6-core backbone with LIUs, pigtails, couplers and patch cords.

12

Media converters

Single-mode Ethernet conversion for remote network links.

VERIFIED CAMERA COUNT

20 cameras total: 15 analog IR + 3 fixed IP + 2 network PTZ.

CONSULTANT-LED DESIGN

Security engineering begins with zones, risks and evidence needs.

D-TECH translates plant operations into camera views, safe routes, transmission paths, recording and response workflows.

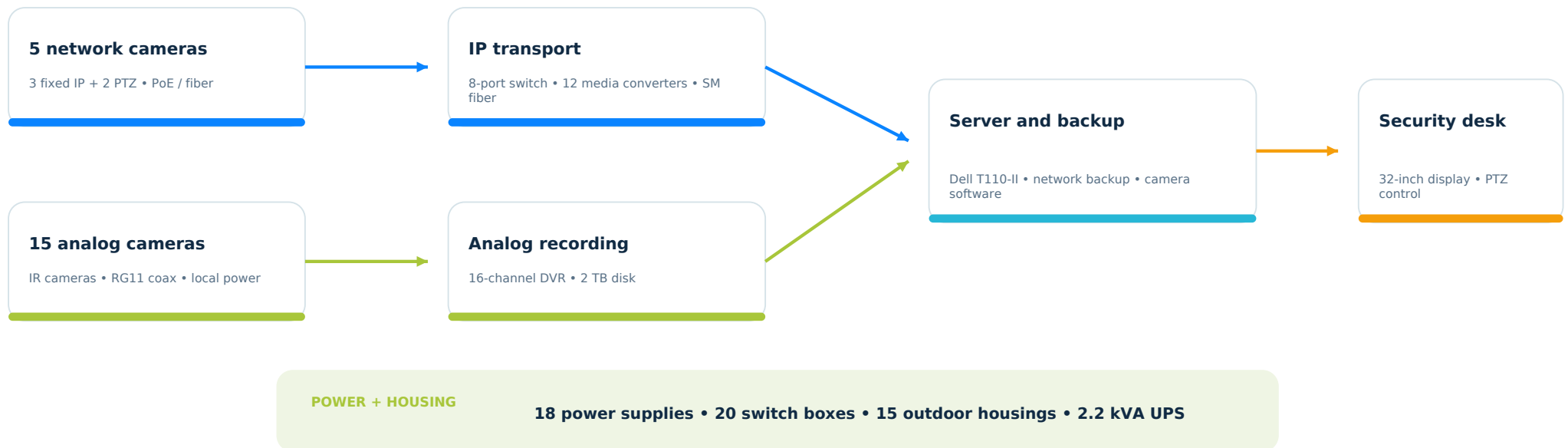


OUTPUT: camera schedule • location drawing • network topology • storage plan • installation method • AMC baseline

SOLUTION DRAWING 01

Hybrid IP + analog surveillance under one operating view.

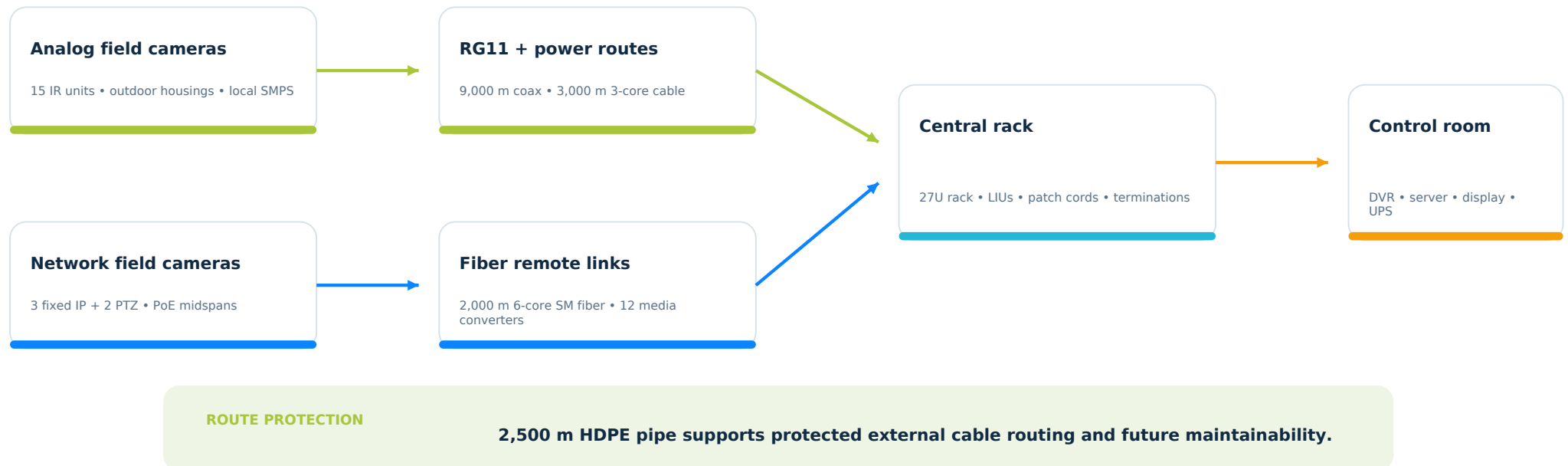
Architecture reconstructed from the 2013 PO components and implementation scope.



SOLUTION DRAWING 02

Long-distance plant connectivity through coax, power and fiber.

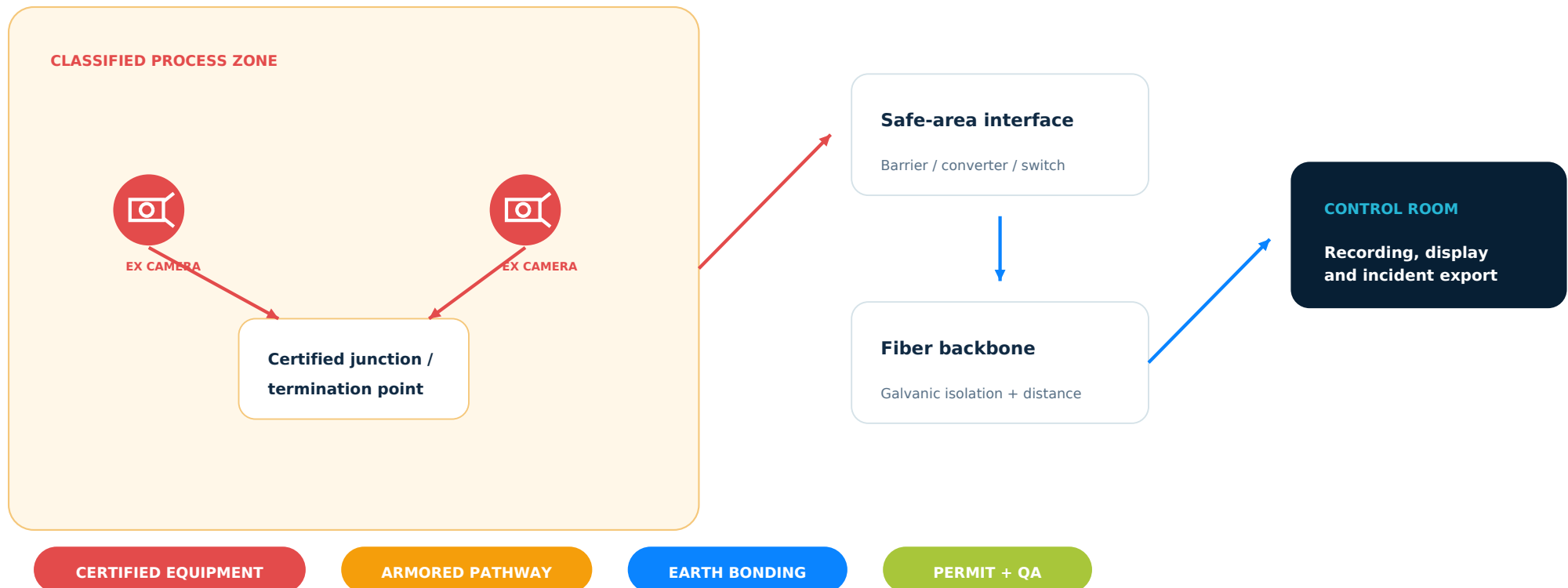
The original BOQ shows a distributed architecture designed to carry camera feeds reliably across the site.



SOLUTION DRAWING 03

Hazardous-area surveillance: visibility with engineered protection.

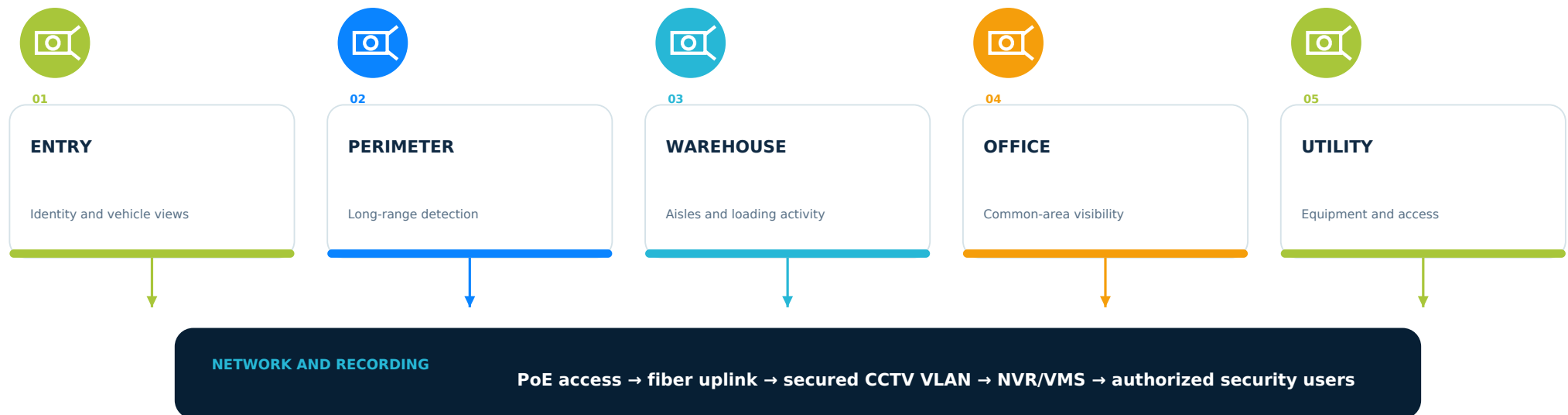
Area classification is an engineering control; final device selection must match the approved zone, gas group and temperature class.



SOLUTION DRAWING 04

Non-hazardous areas: broad coverage with efficient distribution.

Fixed, PTZ and analog cameras can be assigned by view objective, distance, lighting and response requirement.



VERIFIED CONTROL-ROOM SCOPE

The original order delivered the platform behind the cameras.

Recording, server, backup, display, rack and power-protection components supported daily security operations.

16-channel DVR

Standalone recorder supplied with 2 TB hard disk.

Dell T110-II server

Server platform for IP camera management and services.

Network backup

Dedicated backup device for operational resilience.

PTZ controller

AXIS T8310 surveillance control unit for operator use.

32-inch display

HDMI / video monitor for live views and playback.

2.2 kVA UPS

APC UPS supplied and installed with 15-minute backup.

INFRASTRUCTURE

27U central rack • four 4U field racks • LIUs • fiber accessories • CAT6 patch cords

DESIGN TO DELIVERY

A controlled deployment sequence protects safety, quality and uptime.

The PO included installation; D-TECH coordinated pathways, mounting, termination, configuration, testing and handover.

01

Approve drawings

Locations, field of view, routes, power and network ports.

02

Prepare pathways

Supports, conduit, fiber / coax / Cat6 routes and terminations.

03

Install devices

Mounting, glands, seals, junctions, labels and earth bonding.

04

Configure

IP addressing, streams, recording, user rights and display views.

05

Test

Image quality, night view, storage, playback, fail/recovery and exports.

06

Handover

As-built drawings, camera list, credentials and AMC baseline.

QUALITY GATE

No camera is closed until view, recording, labeling and documentation are validated.

SEVEN-YEAR AMC EXPERTISE

Maintenance protects security value long after commissioning.

D-TECH's reported seven-year AMC experience adds preventive care, fault restoration and lifecycle planning.

PREVENT

Lens cleaning • view alignment • enclosure / gland inspection • recording checks

DETECT

Camera-off alerts • storage health • image degradation • network / power faults

RESTORE

Remote triage • site attendance • spares coordination • validated closure

IMPROVE

Recurring-fault review • blind-spot changes • capacity and obsolescence roadmap

DAILY VISIBILITY

MONTHLY HEALTH

QUARTERLY REVIEW

ANNUAL ROADMAP

BUSINESS IMPACT

A documented foundation for plant security and long-term service.

The value comes from verified scope, hybrid-system engineering, structured implementation and lifecycle accountability.

Hybrid visibility

15 analog and 5 network cameras support a unified security objective.

Plant-wide reach

9 km coax, 3 km power cable and 2 km fiber connect distributed locations.

Operator control

DVR, server, display and PTZ controller support live and recorded response.

Infrastructure resilience

Backup, UPS, racks, fiber termination and protected pathways strengthen uptime.

Service continuity

Seven years of D-TECH AMC experience supports prevention and restoration.

Future readiness

Single-mode fiber and network cameras provide a migration path for expansion.

SOURCE + RECORD

PO 3000175931 issued 21 June 2013; seven-year AMC history supplied by D-TECH.